

Shattered Bytes

A Manual Data Carving Serious Game for Computer Forensics Education

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What the game trains

Shattered Bytes focuses on the complete **manual data carving workflow**: detecting meaningful byte patterns, rejecting false leads, reconstructing fragmented evidence, and reporting only what can be defended from the recovered data.

- The player must combine **signature analysis**, **offset reasoning**, and **fragment reconstruction**.
- The campaign avoids a pure “find the header” exercise by adding decoys, partial recovery, XOR obfuscation, and partition-based navigation.
- The final output is both the recovered evidence and a **competency report** on the player's investigative method.

Teaching problem

Data carving can look like simple **signature matching**, but real recovery also requires boundary validation, fragment ordering, decoy rejection, and careful reporting.

Design answer

The player works directly on **raw bytes**, selects exact ranges, reconstructs evidence, verifies the result, and submits a forensic conclusion.

The game turns a low-level forensic task into a sequence of **explicit investigative decisions**.

Operation Shattered Meridian

A regional hospital network has been hit by the Night Meridian ransomware cell. The live breach is contained, but the payment channel, mule identities, ATM cash-out, fraudulent transfer, ransomware payload, and exfiltration path are still hidden in seized devices.

- **Player role:** junior forensic analyst assigned to Agent Root.
- **Mission:** reconstruct six digital evidence items from raw dumps.
- **Final result:** one defensible incident chain.

Core Gameplay Loop

- ① **Identify** the forensic target from the mission briefing.
- ② **Inspect** the dump with search, go, select, and info.
- ③ **Stash** exact byte ranges in the workbench.
- ④ **Compose** fragments, apply XOR when needed, and carve the artifact.
- ⑤ **Verify** the recovered image or text payload in the asset viewer.
- ⑥ **Report**: recovered, partial, or inconclusive.

Design principle

The player is rewarded for a disciplined forensic workflow: observe, validate, reconstruct, and **report only what the bytes support**.

Campaign Progression

Act	Topic	Main forensic skill	Outcome
1	False PNG lead	Header/footer validation	Full recovery
2	Fragmented PNG	Descriptor-based reconstruction	Full recovery
3	Multi-signature triage	Relevance over recognisable files	Full recovery
4	MBR partition	Little-Endian LBA to byte offset	Full recovery
5	Obscured tail	XOR known-plaintext and partial recovery	Partial report
6	Ransomware aftermath	Multi-fragment XOR recovery	Full recovery

- **Hex Editor:** byte-level inspection and range selection.
- **Terminal:** forensic commands such as search, go, select, entropy, xorcalc, xor, and report.
- **Workbench:** fragment storage, ordering, XOR transformation, carving.
- **Evidence Journal:** selected ranges, bad leads, carving attempts.
- **Asset Viewer:** visual/text verification of recovered evidence.
- **Objectives Panel:** explicit investigation goals and manual hints.

Scoring signals

- **exact byte ranges**;
- **false positives** and overlaps;
- extra carving attempts;
- hint usage;
- elapsed time;
- **report correctness**.

Final report

The game summarizes mastery across signature recognition, offset precision, fragment reconstruction, trial-and-error discipline, obfuscation handling, **forensic reasoning**, and reporting quality.

Technical Implementation

- **Frontend:** React and Vite.
- **Levels:** data-driven JSON dumps in `public/levels/`.
- **Generators:** deterministic Python scripts in `scripts/level_generation/`.
- **Evidence:** small source artifacts in `assets/`, runtime images in `public/evidence/`.
- **Runtime:** browser-based interaction, no backend dependency during play.

Reproducibility

The campaign levels can be regenerated with `npm run levels:build`; the project can be built with `npm run build`. This keeps the game **data-driven** and reproducible.

Course alignment

- **file signatures** and data carving;
- fragmentation and unallocated space;
- MBR and **Little-Endian** values;
- anti-forensics and decoys;
- partial recovery and defensible reporting.

Suggested demo

Show Act 1 for **validation**, Act 4 for offset reasoning, Act 5 or 6 for XOR, and close with the **final competency report**.

Questions?